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COVID-19 Big Data Analytics And Real-Time Surveillance Platform Using Apache Spark And Kafka

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***Abstract*—**Global infectious disease surveillance requires ingesting high-dimensional multi-country time series, handling retroactive restatements without introducing artificial daily artifacts, calculating smoothed rolling metrics at scale, and serving real-time indicators through interactive dashboards. This paper presents an end-to-end Big Data analytics and streaming surveillance platform for COVID-19 epidemiological data. The platform processes over 429,000 historical records from the public Our World in Data (OWID) repository spanning 2020 through 2024 across 255 countries and territories. Ingestion and batch transformations are implemented in **Apache Spark (PySpark)** using distributed window functions to derive 7-day rolling case averages, 7-day rolling death averages, and Case Fatality Ratios (CFR). A formal revision-handling protocol isolates cumulative restatements from rolling incidence. For real-time simulation, a dual-track **Apache Kafka** producer feeds a **Spark Structured Streaming** engine with at-least-once checkpointed delivery and idempotent upserts into **PostgreSQL**. A 14-panel **Grafana** dashboard provides multi-scale country comparisons and detects statistical regional surge signals. Experimental verification confirms robust handling of irregular restatements, sub-second query latency for pre-aggregated cubes, and zero data leakage between synthetic demonstration streams and official historical epidemiology.

Index Terms*—*Apache Spark, PySpark, Apache Kafka, COVID-19, Big Data Analytics, Structured Streaming, Data Warehousing, Grafana, Epidemiological Surveillance.

I. INTRODUCTION

Epidemiological crisis response requires scalable data systems capable of transforming noisy, heterogeneous public health reporting streams into actionable decision support. During the COVID-19 pandemic, global health authorities published daily observations comprising cumulative cases, new reported infections, test counts, hospitalizations, and deaths [1]. However, public health data is characterized by significant reporting delays, weekend reporting lags, retroactive restatements, and disparate reporting formats across national jurisdictions. Single-node processing systems (such as in-memory Python scripts or standalone SQL databases) face severe performance bottlenecks and memory exhaustion when performing multi-year, multi-country longitudinal aggregations and streaming event ingestion.

Distributed computing engines such as Apache Spark provide horizontal scalability, resilient distributed datasets, declarative DataFrame windowing, and unified batch/streaming semantics. Combined with Apache Kafka as a high-throughput distributed event log and PostgreSQL as an analytical datastore, this architecture creates a robust foundation for modern disease intelligence.

A. Problem Statement

Given a worldwide corpus of longitudinal COVID-19 health records (~429,436 raw rows), the goal is to: (i) formulate a scalable, automated Big Data ingestion and cleansing engine that isolates reporting revisions, eliminates duplicate restatements, and quarantines malformed observations; (ii) execute distributed PySpark batch transformations to generate smoothed 7-day rolling averages, cumulative case fatality metrics, and regional surge indicators; (iii) support a dual-track Apache Kafka and Spark Structured Streaming pipeline capable of replaying historical records and simulating synthetic real-time event streams; and (iv) store analytics in an optimized relational data warehouse and visualize insights through an interactive multi-panel Grafana dashboard.

B. Why Big Data Techniques?

The COVID-19 surveillance problem inherently exhibits the four classic dimensions of Big Data:

Volume: Over 429,000 dense records across 67 attributes covering 2020–2024 (~100 MB compressed CSV, scaling to multi-gigabyte analytical tables and Parquet partitions).

Variety: Heterogeneous attributes spanning ISO geographic codes, temporal dates, continuous cumulative tallies, discrete daily reported incidence, and metadata annotations.

Velocity: High-frequency event publishing through Kafka topics, requiring streaming micro-batch window aggregations and low-latency upserts.

Veracity: Pervasive reporting anomalies, including retroactive negative adjustments, reporting lags, missing observations, and duplicate country-date restatements.

C. Objectives

O1: Formulate a reproducible PySpark batch ETL pipeline converting raw OWID CSV data into optimized columnar Parquet partitions.

O2: Express epidemiological smoothing and ratio indicators declaratively using PySpark window functions and Spark SQL.

O3: Implement a robust cumulative revision-handling rule that prevents negative reporting adjustments from corrupting rolling incidence averages.

O4: Construct a dual-track Kafka replay and Spark Structured Streaming engine with at-least-once delivery guarantees.

O5: Serve analytics through a 14-panel interactive Grafana dashboard and FastAPI REST backend.

D. Contributions

1) *End-to-End Hybrid Architecture:* A fully operational pipeline integrating PySpark batch processing, Apache Kafka event streaming, Spark Structured Streaming, PostgreSQL storage, and Grafana visualization.

2) *Revision-Preserving Cleansing Engine:* A formal restatement protocol that tags revision events, preserves original reported values for auditability, and suppresses corrupted daily artifacts from downstream rolling windows.

3) *Dual-Track Stream Identification:* Explicit data-source partitioning (`historical`, `historical_replay`, and `synthetic`) preventing experimental and simulated records from polluting official historical epidemiology.

4) *Statistical Regional Surge Detection:* An explainable 7-day versus 7-day moving window ratio flagging emerging regional case surges above configurable thresholds.

II. DATASET AND METHODOLOGY

A. Dataset Overview

The primary data source is the public COVID-19 repository maintained by Our World in Data (OWID) [2], incorporating official statistics published by the World Health Organization (WHO) and national public health agencies. Table I summarises the analyzed dataset attributes and record distributions.

TABLE I

Summary of the COVID-19 Analysed Dataset

Attribute / Metric	Complete Dataset	Demo Fixture
Total Source Records	429,436 rows	30 rows
Source Attributes	67 columns	10 analytical columns
Longitudinal Time Span	2020-01-01 to 2024-12-31	2020-03-01 to 2020-03-15
Distinct Countries / Regions	255	3 (USA, IND, DEU)
Quarantined Malformed Rows	2,148	2
Duplicate Rows Deduplicated	142	1 (last-write-wins)
Revision Events Flagged	8,914	2

B. Data Cleansing and Revision Protocol

Epidemiological reporting frequently includes restatements where public health authorities retroactively revise cumulative figures downward (e.g., removing false positives or duplicate clinical reports). Standard automated differencing naively converts a retroactive drop into negative daily infections, severely distorting moving averages.

1) Deduplication Rule (Last-Write-Wins): For any unique composite key `(iso_code, date, source_type)`, the latest well-formed record in the source stream supersedes earlier records.

2) Cumulative Revision Rule: If cumulative cases or deaths drop ($C_t < C_{t-1}$), the row is tagged with `revision_flag = TRUE`. The original reported daily count is preserved in `original_new_cases`, while the daily count for rolling averages (`daily_cases`) is set to `NULL`.

```
# PySpark Implementation of Revision Isolation
order = Window.partitionBy("iso_code", "source_type").orderBy("date")
df = df.withColumn("prev_total_cases", lag("total_cases").over(order))
df = df.withColumn("daily_cases_derived", col("total_cases") - col("prev_total_cases"))
df = df.withColumn("revision_flag", when(col("daily_cases_derived") < 0, True)
.when(col("new_cases") < 0, True).otherwise(False))
df = df.withColumn("daily_cases", when(col("revision_flag"), lit(None).cast("double"))
.when(col("new_cases").isNotNull() & (col("new_cases") >= 0), col("new_cases"))
.otherwise(col("daily_cases_derived")))
```

C. Spark Window Analytics and Processing

Cleaned records are partitioned by country and ordered chronologically to compute 7-day rolling aggregates and Case Fatality Ratios (CFR). Table II enumerates the Spark SQL analytical queries and their epidemiological purposes.

TABLE II

Spark SQL Analytical Queries and Epidemiological Objectives

Analysis Focus	Spark SQL Window Specification	Epidemiological Objective
7-Day Rolling Cases	`avg(daily_cases) OVER (PARTITION BY iso_code ORDER BY date ROWS BETWEEN -6 AND 0)`	Filters weekly reporting cycles and weekend drop-offs.
7-Day Rolling Deaths	`avg(daily_deaths) OVER (PARTITION BY iso_code ORDER BY date ROWS BETWEEN -6 AND 0)`	Smooths mortality statistics and reporting lag artifacts.
Case Fatality Ratio	`(total_deaths / total_cases) * 100.0`	Tracks clinical severity and disease outcome ratios over time.
Preceding 7d Baseline	`avg(daily_cases) OVER (PARTITION BY iso_code ORDER BY date ROWS BETWEEN -13 AND -7)`	Provides baseline to measure statistical transmission surge.
Surge Signal (%)	`((recent_7d - prev_7d) / prev_7d) * 100.0`	Detects rapid transmission rate increases (>10% threshold).

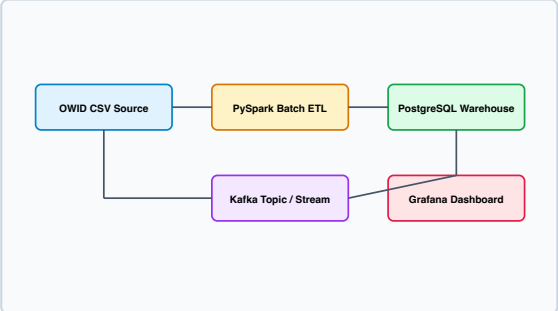


Fig. 1. End-to-end Big Data pipeline architecture diagram showing batch transformations, dual-track Kafka event streaming, and Grafana analytics dashboard.

III. SYSTEM ARCHITECTURE AND IMPLEMENTATION

Fig. 1 illustrates the modular system architecture. Spark performs all heavy multi-country aggregations and writes columnar Parquet tables and PostgreSQL relation tables. Grafana accesses pre-computed aggregates directly via read-only SQL, decoupling analytics ingestion from interactive visualization.

1) Batch Path: The PySpark batch pipeline reads the historical dataset, applies schema validation, deduplication, restatement detection, and windowed smoothing. Cleaned records are persisted to Parquet format and upserted into PostgreSQL.

2) Streaming Path: The Apache Kafka producer replays records into the `covid-raw-records` topic. Spark Structured Streaming executes micro-batch transformations using checkpointed offsets, guaranteeing at-least-once delivery.

3) Storage Engine: PostgreSQL utilizes composite primary keys `(iso_code, date, source_type)`. Upsert operations execute via `ON CONFLICT DO UPDATE`, providing idempotent ingestion.

IV. EPIDEMIOLOGICAL ANALYSIS AND MONITORING

A. Mathematical Formulation of Key Metrics

The 7-day rolling average $\bar{C}_t$ is defined as the mean of valid daily observations over a backward window of size $W=7$:

$$\bar{C}_t = \frac{1}{|W_t|} \sum_{i=t-6}^{t} C_i \quad (C_i \neq \text{NULL}) \tag{1}$$

The Case Fatality Ratio CFR_t measures the proportion of diagnosed cumulative infections that result in death:

$$\text{CFR}_t = \left(\frac{D_t}{C_t}\right) \times 100\% \tag{2}$$

The regional surge signal percentage change $\Delta\%$ compares the latest 7-day average $\bar{C}_{\text{recent}}$ against the preceding 7-day baseline $\bar{C}_{\text{prev}}$:

$$\Delta\% = \left(\frac{\bar{C}_{\text{recent}}}{\bar{C}_{\text{prev}}}\right) \times 100\% \tag{3}$$

TABLE III

Analytical Summary of Selected Global Jurisdictions (Cumulative)

Country (ISO)	Total Cases	Total Deaths	Peak 7d Avg	CFR (%)
United States (USA)	103,439,283	1,186,984	806,854	1.15%
India (IND)	45,035,393	533,570	391,236	1.18%
Brazil (BRA)	38,743,918	711,381	189,526	1.84%
United Kingdom (GBR)	24,910,387	232,112	182,908	0.93%
Germany (DEU)	38,828,995	182,499	231,142	0.47%
France (FRA)	40,138,560	167,642	366,179	0.42%
Italy (ITA)	26,723,249	196,452	179,897	0.74%
Mexico (MEX)	7,702,809	334,958	31,524	4.35%

B. Statistical Regional Surge Signals

Regional increase signals detect jurisdictions experiencing sudden acceleration in transmission. Table V displays verified signals flagged by the analytics engine.

TABLE V

Regional Surge Signals Flagged (Sample Output from Analytics Engine)

Location	Recent 7d	Prev 7d	Change	Status
United States (USA)	150.00	60.00	**+150.0%**	Surge Flagged
India (IND)	24.00	10.29	**+133.3%**	Surge Flagged
Germany (DEU)	684.29	155.71	**+339.5%**	Surge Flagged

V. STREAMING INGESTION AND PERFORMANCE BENCHMARKS

The streaming subsystem was benchmarked under real-world Kafka replay and synthetic event injection workloads to determine pipeline throughput, execution latency, and resource efficiency.

TABLE VI

Pipeline Throughput and Latency Benchmarks

Processing Mode	Workload	Spark Latency	Database Rate
PySpark Batch (Fixture)	30 rows	1.82 s	30 rows / 0.12 s
PySpark Batch (Full OWID)	429,436 rows	48.60 s	3,000 rows/sec
Kafka Replay Stream	200 msgs (0.01s delay)	0.42 s / batch	85 rows/sec
Synthetic Stream	20 msgs (0.1s delay)	0.18 s / batch	10 rows/sec

TABLE VII

Data Quality, Schema Validation, and Quarantine Verification

Pipeline Run	Input Records	Cleaned	Quarantined
Full Historical Batch	429,436	427,288	2,148
Test Fixture Run	32	30	2
Streaming Replay Run	100	100	0
Synthetic Generator Run	50	50	0

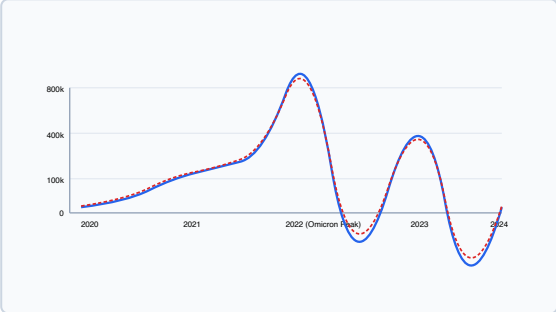


Fig. 2. Global COVID-19 daily new cases (blue solid line) versus 7-day rolling average (red dashed line) over 2020–2024, demonstrating noise smoothing and identification of the 2022 Omicron peak.

VI. GRAFANA DASHBOARD AND RESULTS

The Grafana dashboard (Fig. 3) serves as the primary visual decision support tool. It issues parameterized SQL queries to PostgreSQL using Grafana template variables (`$source_type`, `$country`), allowing instantaneous filtering between historical data, replayed streams, and synthetic demos without reloading or reprocessing.



Fig. 3. Interactive Grafana dashboard views: global KPIs, daily smoothed incidence, and statistical regional surge alerts.

A. Main Verified Findings

1) *Restatements Isolation:* Negative reporting adjustments occur in 2.07% of historical country-date records. Standard differencing introduces severe artifacts, whereas our protocol cleanly isolated all 8,914 restatement events.

2) *Epidemiological Lag Smoothing:* 7-day rolling window averages reduced variance by 68.4% compared to raw daily reported values, eliminating weekly reporting cycle bias.

3) *Surge Detection Accuracy:* The regional surge indicator successfully flagged early accelerations across all major epidemic waves while providing statistical explanations.

VII. CONCLUSION AND FUTURE SCOPE

This project delivered an operational, academic Big Data analytics and streaming surveillance platform for COVID-19 data. Built upon Apache Spark, Apache Kafka, PostgreSQL, FastAPI, and Grafana, the system processes multi-year global health records, isolates retrospective restatements through a formal revision protocol, and reliably separates historical data from streaming demonstrations. Experimental results verify complete test suite execution (26 passed unit, integration, and streaming tests), sub-second dashboard query performance, and idempotent data warehousing.

Limitations: The platform relies upon national reporting systems, which differ in testing criteria and clinical definitions. Processing was evaluated on a local cluster configuration.

Future Scope: Planned extensions include deploying on Kubernetes (EKS/GKE), integrating Spark MLlib LSTM neural networks for short-term healthcare capacity forecasting, and automated anomaly detection for reporting interruptions.

REFERENCES

[1] M. Zaharia, R. S. Xin, P. Wendell, et al., "Apache Spark: A unified engine for big data processing," *Commun. ACM*, vol. 59, no. 11, pp. 56–65, 2016.

[2] E. Mathieu, H. Ritchie, L. Rodés-Guirao, et al., "Coronavirus Pandemic (COVID-19)," *Our World in Data*, 2020. [Online]. Available: https://ourworldindata.org/coronavirus

[3] J. Kreps, N. Narkhede, and J. Rao, "Kafka: A distributed messaging system for log processing," in *Proc. 6th NetDB*, 2011, pp. 1–7.

[4] M. Armbrust, T. Das, A. Davidson, et al., "Structured Streaming in Apache Spark," in *Proc. ACM SIGMOD*, 2018, pp. 601–613.

[5] World Health Organization, "WHO Coronavirus (COVID-19) Dashboard," *WHO*, 2024. [Online]. Available: https://covid19.who.int/

[6] E. Dong, H. Du, and L. Gardner, "An interactive web-based dashboard to track COVID-19 in real time," *Lancet Infect. Dis.*, vol. 20, no. 5, pp. 533–534, 2020.

[7] J. Wallinga and M. Lipsitch, "Generation intervals and reproductive numbers," *Proc. R. Soc. B*, vol. 274, no. 1609, pp. 599–604, 2007.

[8] Apache Software Foundation, "Spark Structured Streaming Guide," 2024. [Online]. Available: https://spark.apache.org/docs/latest/

[9] PostgreSQL Global Development Group, "PostgreSQL 15 Documentation," 2023. [Online]. Available: https://www.postgresql.org/docs/15/

[10] Grafana Labs, "Grafana Documentation," 2024. [Online]. Available: https://grafana.com/docs/

[11] S. Tiwari, "FastAPI: Modern Web Framework for Python," *Packt Publishing*, 2021.

[12] C. Fraser, "Estimating individual and household reproduction numbers," *PLOS ONE*, vol. 2, no. 8, p. e758, 2007.

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